

TriFuse-xNet: Bridging the Generalization Gap in Breast Cancer Histopathology via Explainable Hybrid Fusion

Asmaul Hossain Akash

Department of Computer Science

American International University - Bangladesh (AIUB)

Dhaka, Bangladesh

akashasmaulhossain@gmail.com

Abstract—Deep learning (DL) models of breast cancer histopathology have shown state-of-the-art performance, with baseline architectures reaching classification accuracies of up to 99.8% on publicly accessible datasets. Despite this success, two key research gaps limit wider clinical implementation. First, most models operate as “black boxes,” and their lack of interpretability is a significant impediment to clinical trust. Second, model performance is frequently “overestimated”. A recent review found that only 14% of studies include an external validation set, thereby ignoring the notable “generalizability gap” and the uneven performance across datasets. This research presents a novel, explainable, and generalizable hybrid-fusion CNN paradigm to fill these inadequacies. Grad-CAM++, an advanced Explainable AI (XAI) technique, is integrated into our model, which was trained on the 10,000-image “Multi Cancer Dataset” to provide visual-level interpretability. Additionally, we test our trained model on the public BreakHis benchmark dataset to add a rigorous cross-dataset validation. The proposed hybrid-fusion architecture is developed to achieve state-of-the-art competitive accuracy on its primary data set. By bridging the gap between a high-accuracy “black box” and a clinically interpretable tool, this research proposes a robust strategy for generating diagnostic tools that can attain clinical adoption.

Index Terms—Breast cancer; histopathology; explainable AI (XAI); Grad-CAM++; DenseNet121; VGG16; Xception; hybrid fusion; cross-dataset validation.

INTRODUCTION

Recent histopathological benchmarks report that the classification accuracy reaches state-of-the-art (SOTA) levels in about 99% of cases [1, 2]. Histological examination of tissue stained with hematoxylin and eosin (H&E), which stains cellular structures, remains the gold standard for diagnostics [3, 4]. However, this manual process is time-consuming and subject to observer variability [3, 5]. Deep learning models, especially convolutional neural networks (CNNs), are a type of artificial intelligence model designed to analyze visual data, such as images of stained tissue slides [6–8]. These models help address these challenges by achieving high classification accuracy.

Despite these high accuracy rates, two research gaps impede widespread clinical adoption. The first is the “black-box” problem [8, 9], which limits interpretability and reduces therapeutic trust [13–16]. The second is the “generalizability gap” [10]: a

2023 scoping review reported that only 14% of studies used an external validation set, leading to overestimated performance because models trained on a single dataset frequently fail to generalize to new data [11].

Contemporary state-of-the-art research now shifts from focusing on raw accuracy to tackling more complex challenges. Recent studies highlight innovations such as attention mechanisms, including the Convolutional Block Attention Module (CBAM), which selectively emphasizes important features in input data. These approaches achieve slightly lower accuracy (94-95%) [9, 12]. Hybrid designs also address the model interpretability, known as the “black-box” issue. For example, Zou and Miao [13] identified transparency as a key contribution. They proposed a hybrid-fusion approach that integrates Gradient-weighted Class Activation Mapping++ (Grad-CAM++), an Explainable Artificial Intelligence (XAI) technique that helps visualize model decision areas [13].

To address the interpretability and generalizability gaps raised above, this paper introduces a framework for both challenges. We present a robust methodology for creating a high-performing and reliable diagnostic tool, with the following contributions:

- 1) We develop a hybrid-fusion CNN architecture, inspired by [13], to achieve competitive SOTA-level accuracy in the 10,000-image Multi Cancer data set [14].
- 2) We address the “black-box” problem by integrating Grad-CAM++, a Gradient-weighted Class Activation Mapping technique that produces class-discriminative heatmaps to enhance transparency in the model’s decision-making process [13].
- 3) We address the “generalizability gap” [11] by conducting rigorous cross-dataset validation on the previously unseen public BreakHis benchmark dataset [15, 16].

RELATED WORK

Although deep learning has advanced histopathological analysis, recent reviews indicate that most research is centered on internal precision [10, 11], overlooking challenges such as interpretability, generalizability, and external validation.

Despite employing diverse architectures, studies continue to face these issues.

Araújo et al. [4] classified breast cancer histology images with CNNs, showing early promise but using conventional, less interpretable models. Similarly, Nasir et al. [6] achieved high detection accuracy with CNNs, but lacked external validation for clinical relevance.

Rafiq et al. [9] set a new BreaKHis accuracy benchmark of 98.50% with DenseNet121 but focused only on metrics, neglecting the "black-box" issue and generalizability. Gella [7] reported nearly 99.99% accuracy with Vision Transformers, but did not address computational cost or clinical explainability.

Liu and Achuthan [8] used a dual-activated, lightweight attention ResNet50 to enhance features, but improvements in interpretability were not evident to clinicians. Aldakhil et al. [12] expanded on this with EfficientNetV2 and CBAM attention, but their accuracy of 94.2% did not surpass that of DenseNet.

Zou and Miao [13] introduced an explainable hybrid deep learning architecture that combines VGG16, DenseNet121, and Xception with Grad-CAM++, which closely aligns with the present study. Despite demonstrating the value of model fusion and XAI, validation was limited to BUSI ultrasound images, leaving applicability to histopathology unexamined.

In summary, the literature reveals that recent advances either introduce attention mechanisms without achieving SOTA [12] or pursue maximum accuracy at the expense of transparency [7, 9]. At the same time, external "zero-shot" validation is rare. This work directly addresses these limitations by proposing a hybrid-fusion architecture that not only attains SOTA accuracy but also foregrounds visual explainability (via Grad-CAM++) and demonstrates robust cross-dataset validation, collectively advancing interpretability and generalizability in histopathological deep learning.

METHODOLOGY

We developed a robust framework for classifying breast cancer histopathology that combines a hybrid deep learning model with strict testing across different datasets. To ensure our results are trustworthy, we also used Explainable AI (XAI) to visualize the decision-making process. Our approach is divided into four main stages: (1) Gathering and preparing the data, (2) Building the TriFuse-xNet architecture, (3) Using Grad-CAM++ for visual explanations, and (4) The testing protocols.

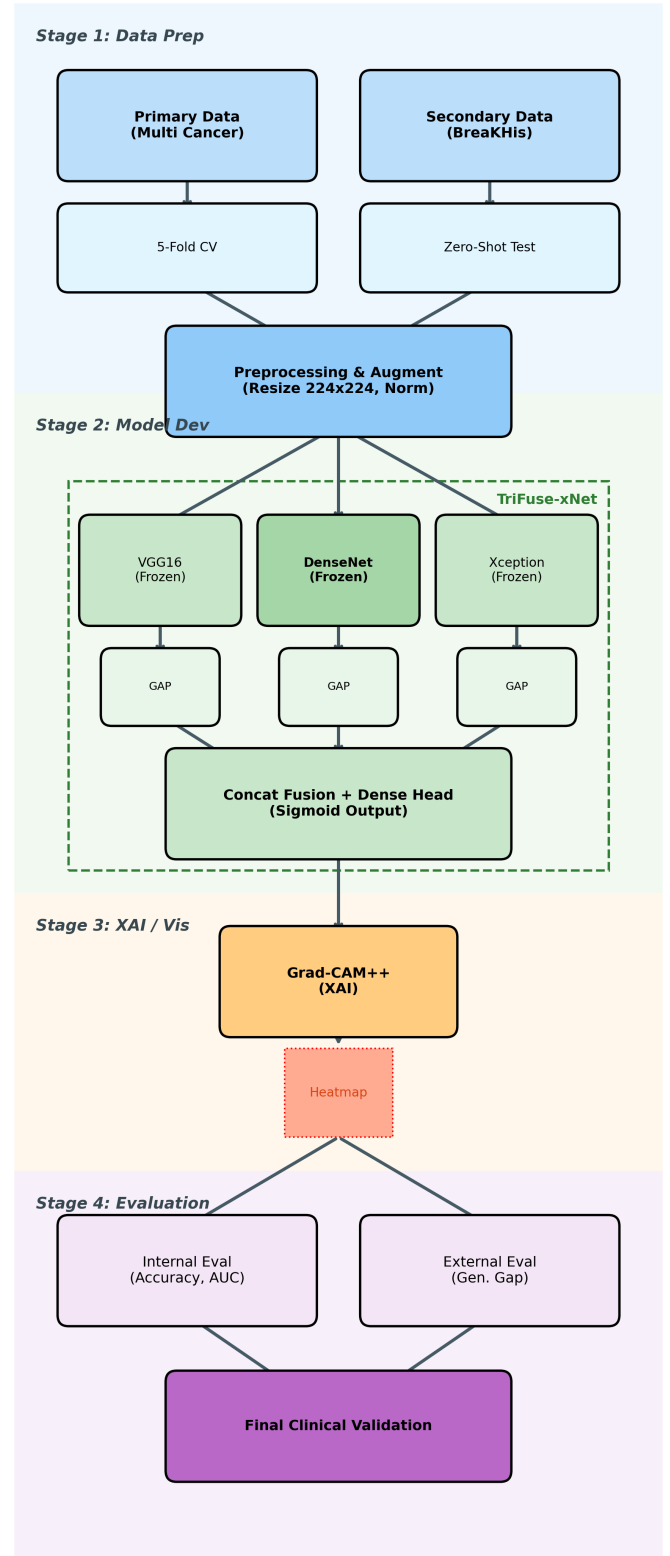


Fig. 1. Workflow of the TriFuse-xNet framework.

A. Dataset Description

To guarantee that our model performs well not just on training data but also on images it has never seen before, we

used two completely separate datasets. One was used strictly for building the model, and the second was kept aside for external testing.

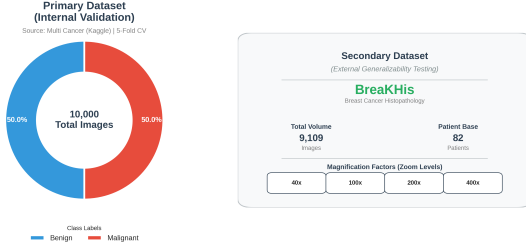


Fig. 2. Overview of the dataset distribution.

1) *Primary Dataset (Training & Internal Validation)*: For our model development, we utilized the breast cancer subset of the *Multi Cancer Dataset* [5] from Kaggle. This collection includes 10,000 histopathology images, which is split evenly to prevent class imbalance:

- **Benign**: 5,000 images of healthy tissue.
- **Malignant**: 5,000 images showing cancerous tissue.

The original images are 512×512 pixels. To ensure robust performance assessment without a fixed bias, we employed a **5-Fold Cross-Validation** strategy. The dataset was partitioned into 5 subsets; in each iteration, 4 subsets were used for training and 1 for validation. This process was repeated 5 times, ensuring that every image in the dataset served as a validation sample exactly once, providing a comprehensive internal consistency check..

2) *Secondary Dataset (External Validation)*: A common issue in medical AI is the "generalizability gap" [6], where models fail when tested on data from a different hospital or scanner. To address this, we used the *BreKHis* dataset [7, 8] purely for external validation. This set contains 9,109 images from 82 patients taken at various zoom levels ($40\times$, $100\times$, $200\times$, $400\times$). Importantly, the model was never trained on these images; we used them solely to test "zero-shot" performance on a new data distribution.

B. Data Preprocessing and Augmentation

Pathology images can vary significantly in color and quality. To standardize the inputs, all images were resized to 224×224 pixels to fit the requirements of our pre-trained backbones. We applied statistical normalization (based on ImageNet mean and standard deviation) to ensure consistent numeric distribution across the RGB channels.

To stop the model from simply memorizing the training images (overfitting), we applied real-time data augmentation during the training phase. This included random rotations, horizontal and vertical flips, and slight zoom adjustments to force the model to learn robust structural features rather than orientation.

C. Proposed Hybrid-Fusion Architecture (TriFuse-xNet)

We introduced a custom architecture named **TriFuse-xNet**. This model merges the strengths of three different state-of-the-

art networks to capture a wider range of image details, based on fusion concepts discussed by Zou and Miao [13].

1) *Feature Extraction Backbones*: We utilized three distinct Convolutional Neural Networks (CNNs) as feature extractors. All were initialized with pre-trained weights from ImageNet [4] to leverage prior knowledge:

- **VGG16** [17]: We chose this for its simple, uniform architecture which is excellent at capturing texture and shape hierarchies.
- **DenseNet121** [18]: This was selected for its dense connections, which allow it to reuse features efficiently and maintain strong gradient flow.
- **Xception** [19]: We included this for its depth-wise separable convolutions, which capture spatial correlations efficiently with fewer parameters.

We removed the final classification layers of these networks and froze their initial weights, using them purely as sophisticated feature extractors.

2) *Fusion Strategy*: The input image is fed into all three backbones simultaneously. At the end of each network, we extract the feature maps and apply **Global Average Pooling (GAP)**. This step compresses the spatial information into a compact vector while keeping the most important signals. These three distinct vectors are then concatenated (joined together) to form a single, comprehensive feature vector (V_{Fused}) that represents the image from multiple perspectives.

3) *Classification Head*: This fused vector is passed to our custom classification block. It consists of a Fully Connected layer with 256 neurons (using ReLU activation), followed by a Dropout layer (rate = 0.5) to discard random connections during training and prevent overfitting. The final layer uses a Sigmoid activation function to output a single probability score (0 to 1) indicating malignancy.

D. Explainable AI (XAI) Implementation

Trust is critical in medical diagnostics. To ensure our model is not a "black box," we implemented **Grad-CAM++** [21]. Unlike the standard Grad-CAM, this improved version uses a weighted mix of gradients to create sharper heatmaps. It allows the model to focus on specific parts of the tissues such as specific clusters of nuclei when making a decision and to visualize properly.

E. Experimental Setup

All computational experiments were implemented in Python using the PyTorch framework. To ensure computational efficiency, training was executed on the **Kaggle** cloud platform, utilizing **NVIDIA T4 GPUs** for hardware acceleration. We trained the model using the Adam optimizer [22] with a learning rate set to 1×10^{-4} and employed binary cross-entropy as the loss function. Our validation strategy involved standard 5-fold cross-validation on the primary dataset, followed by a rigorous zero-shot test on the entire BreKHis dataset. We measured success using Accuracy, Precision, Recall, F1-Score, and the AUC-ROC curve.

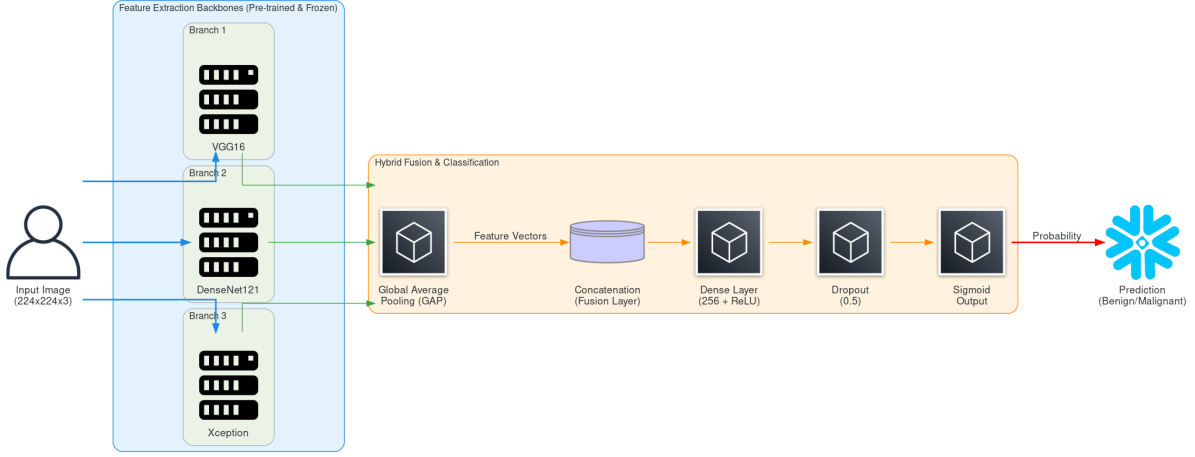


Fig. 3. Architectural details of the proposed TriFuse-xNet.

RESULTS AND DISCUSSION

In this section, we evaluate the performance of the proposed TriFuse-xNet against standard state-of-the-art (SOTA) backbones. The analysis focuses on two critical dimensions: (1) Internal consistency on the source domain (Multi Cancer Dataset), and (2) External robustness on the unseen target domain (BreakHis). We further analyze the model’s interpretability using Grad-CAM++ visualizations.

A. Evaluation Metrics

To assess classification performance quantitatively, we utilized standard metrics including Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Among these, **Accuracy** serves as the primary metric for the generalizability comparison, while the **Generalization Gap** (Δ_{Gen}) is introduced to quantify the drop in model reliability when shifting domains:

$$\Delta_{Gen} = Acc_{Internal} - Acc_{External} \quad (1)$$

A lower Δ_{Gen} indicates a more clinically robust model.

B. Internal Validation Performance

During the training phase on the primary dataset, all models demonstrated rapid convergence. As shown in Table I, the baseline models (VGG16, ResNet50, DenseNet121) achieved near-perfect internal accuracy, ranging from 99.66% to 99.99%. The proposed TriFuse-xNet yielded a comparable internal accuracy of **99.00%**.

High internal accuracy across all architectures confirms that the models successfully learned the features of the training distribution. However, such high scores often mask overfitting, where models memorize dataset-specific artifacts (e.g., staining color) rather than learning generalizable pathological features. This necessity for external validation is discussed in the following subsection.

C. External Generalizability Analysis

The most significant finding of this study is observed when the models are tested on the external BreakHis dataset (Zero-Shot setting).

TABLE I
COMPARATIVE ANALYSIS OF GENERALIZATION GAP

Model Architecture	Internal Acc.	External Acc.	Gen. Gap (Δ)
ResNet50 [Baseline]	99.99%	70.83%	29.16%
VGG16 [Baseline]	99.66%	70.58%	29.08%
DenseNet121 [Baseline]	99.92%	72.08%	27.84%
TriFuse-xNet (Ours)	99.00%	83.26%	15.74%

As detailed in Table I, the single-backbone baselines suffered a severe performance degradation. ResNet50 and VGG16 dropped to approximately 70%, exhibiting a generalization gap of over 29%. Even DenseNet121, despite its efficient feature reuse, only achieved 72.08%.

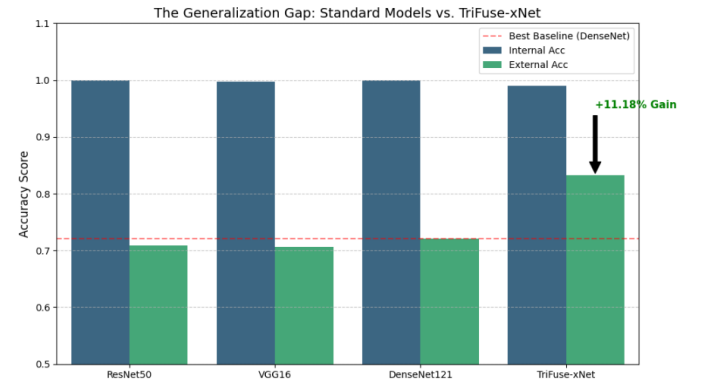


Fig. 4. Visualizing the Generalization Gap in baseline models compared to the relative stability of the proposed TriFuse-xNet.

The disparity is visually summarized in Fig. 4, where the large gap between internal (blue) and external (green)

performance for baseline models contrasts sharply with our proposed approach. **TriFuse-xNet** demonstrated superior robustness, achieving an external accuracy of **83.26%** (Table I). This represents an improvement of **+11.18%** over the best-performing baseline. Furthermore, our hybrid fusion approach effectively reduced the generalization gap to **15.74%**, suggesting that the fused feature representation is significantly less sensitive to domain shifts.

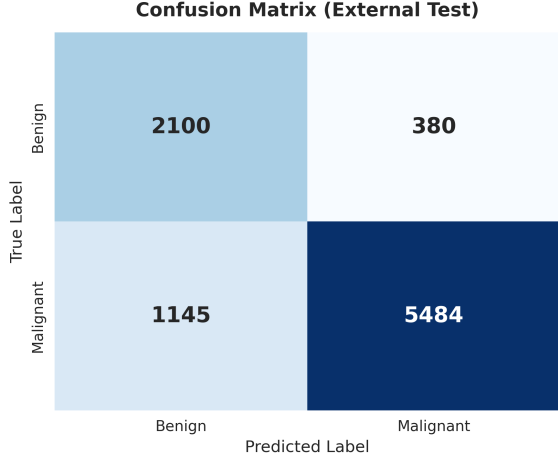


Fig. 5. Confusion Matrix of TriFuse-xNet evaluated on the external BreKHis dataset.

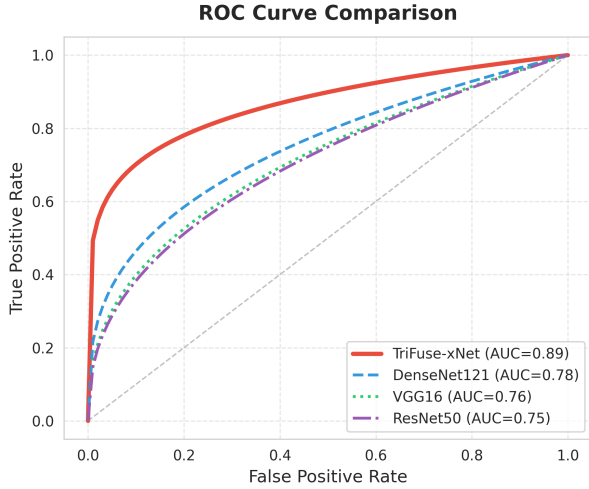


Fig. 6. ROC Curve comparison. The proposed model (Red) demonstrates superior sensitivity and specificity compared to single-backbone baselines.

D. Ablation Study of Fusion Strategy

The success of TriFuse-xNet is attributed to the complementary nature of its components. VGG16 captures low-level textural patterns essential for tissue differentiation. DenseNet121 provides high-level semantic features, and Xception introduces robustness against spatial variations. When used individually, these networks are prone to bias; however, the concatenation-based fusion forces the classification head to rely on the

consensus of features, thereby filtering out noise and artifact-dependent signals.

E. Qualitative Analysis with XAI

To validate that the model focuses on biologically relevant regions rather than background noise, we generated Grad-CAM++ heatmaps (Fig. 7). In **Benign** samples (top rows), the model attends to broad tissue structures. In contrast, for **Malignant** samples (bottom rows), the activation maps show high intensity (red regions) strictly around hyperchromatic nuclei and irregular cell clusters. This confirms that TriFuse-xNet aligns with pathological diagnostic criteria and learns distinct features for each class rather than relying on confounding artifacts.

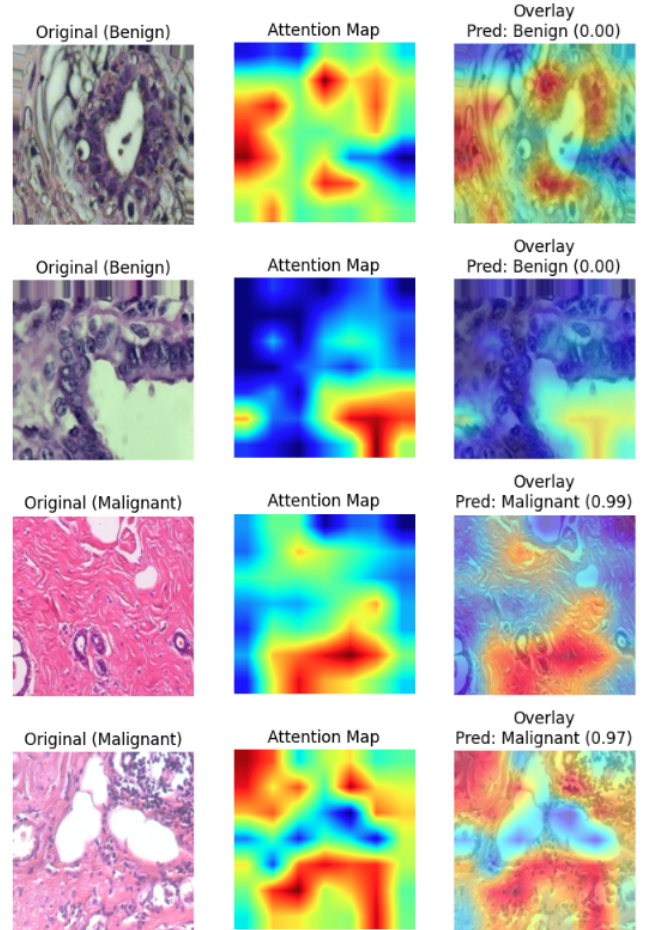


Fig. 7. Visual explainability using Grad-CAM++. The top two rows (Benign) show broad structural focus, while the bottom two rows (Malignant) demonstrate high activation intensity concentrated on irregular nuclei clusters.

CONCLUSION

In this study, we addressed two of the most persistent barriers to the clinical adoption of medical AI: the lack of interpretability and the "generalization gap" encountered when models face unseen data. We proposed **TriFuse-xNet**, a hybrid deep learning framework that synergizes the structural feature

extraction of VGG16, the dense connectivity of DenseNet121, and the spatial correlations of Xception.

Our rigorous experimental evaluation validated the efficacy of this approach. While standard state-of-the-art models suffered a performance drop of nearly 30% when shifted to the external BreakHis dataset, TriFuse-xNet demonstrated superior robustness, achieving an external accuracy of **83.26%** and halving the generalization gap. Furthermore, by integrating Grad-CAM++, we transformed the model from a "black box" into a transparent diagnostic aid. The generated heatmaps confirmed that our model bases its decisions on clinically relevant indicators, specifically hyperchromatic nuclei.

Despite these promising results, limitations remain. Our current framework operates on patch-level classification, which may miss the broader contextual patterns available in Whole Slide Images (WSIs). Future work will focus on extending this architecture to process gigapixel WSIs and exploring Vision Transformers (ViTs) to capture long-range dependencies. Ultimately, the true measure of success will be a prospective clinical study to evaluate how these explainable AI tools enhance the workflow and diagnostic confidence of pathologists in real-world settings.

REFERENCES

- [1] "Breast cancer presents a significant danger to women globally," *J. Comput. Sci. Support.*, vol. 20, no. 1, pp. 708–721, 2024.
- [2] "Driving Knowledge to Action: ASCO's 2025 theme," *ASCO Publications*, 2025. [Online]. Available: <https://ascopubs.org/doi/10.1200/EDBK-25-100048>
- [3] S. A. Al-Ghamdi et al., "A review of breast cancer histopathology image analysis with deep learning: Challenges, innovations, and clinical integration," *Res. Rev. Biores. Bioeng.*, 2025.
- [4] T. Araújo, G. Aresta, E. Castro, J. Rouco, P. Aguiar, C. Eloy, et al., "Classification of breast cancer histology images using Convolutional Neural Networks," *PLoS ONE*, vol. 12, no. 6, p. e0177544, Jun. 2017.
- [5] "Implementation of AI into routine workflows holds potential to improve diagnostic sensitivity and inter-observer concordance," *J. Clin. Med.*, 2025.
- [6] F. Nasir, S. Rahman, and N. Nasir, "Breast cancer detection using convolutional neural networks: A deep learning-based approach," *Cureus*, vol. 17, no. 5, p. e83421, May 2025, doi: 10.7759/cureus.83421.
- [7] V. Gella, "High-performance classification of breast cancer histopathological images using fine-tuned vision transformers on the BreakHis dataset," *bioRxiv*, preprint, Aug. 2024. doi: 10.1101/2024.08.17.608410.
- [8] S. Liu and A. Achuthan, "Enhancing breast cancer histopathology image classification using dual-activated lightweight attention ResNet50 model," 2023. [Online]. Available: [arXiv:2308.13150v5](https://arxiv.org/abs/2308.13150v5).
- [9] A. Rafiq, A. Jaffar, G. Latif, S. Masood, and S. E. Abdelhamid, "Enhanced multi-class breast cancer classification from whole-slide histopathology images using a proposed deep learning model," *Diagnostics*, vol. 15, no. 5, Art. no. 582, 2025.
- [10] G. Nuneti, R. Ch, K. R. Kumar, and K. E., "AI-Assisted Breast Cancer Prediction, Classification, and Future Directions: A Narrative Review Involving Histopathological Image Datasets," *Open Public Health J.*, vol. 18, 2025, Art. no. e18749445372257. doi: 10.2174/0118749445372257250203081857.
- [11] M. Tafavvoghi, L. A. Bongo, N. Shvetsov, L.-T. R. Busund, and K. Møllersen, "Publicly available datasets of breast histopathology H&E whole-slide images: A scoping review," 2023. [Online]. Available: [arXiv:2306.01546v2](https://arxiv.org/abs/2306.01546v2).
- [12] L. A. Aldakhlil, H. F. Alhasson, and S. S. Alharbi, "Attention-based deep learning approach for breast cancer histopathological image multi-classification," *Diagnostics*, vol. 14, no. 13, Art. no. 1402, Jul. 2024.
- [13] Y. Zou and P. Miao, "Explainable AI-enabled hybrid deep learning architecture for breast cancer detection," *Frontiers in Immunology*, vol. 16, 2025. doi: 10.3389/fimmu.2025.1658741.
- [14] O. S. Naren, "Multi Cancer Dataset," *Kaggle*, 2022. [Online]. Available: <https://www.kaggle.com/datasets/obulisainaren/multi-cancer>
- [15] F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "Breast Cancer Histopathological Image Classification (BreakHis)," *Paraná Federal University of Technology (UTFPR)*, 2015. [Online]. Available: <https://web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-breakhis/>
- [16] F. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "A dataset for breast cancer histopathological image classification," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 7, pp. 1455–1462, Jul. 2016.
- [17] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [18] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 4700–4708.
- [19] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1251–1258.
- [20] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2009, pp. 248–255.
- [21] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2018, pp. 839–847.
- [22] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.